

Yash Dagade

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EDUCATION

Duke University

Durham, NC

B.S. in Mathematics and Philosophy (A.B. *Duke Full-Ride Merit Scholar*)

2024 – 2028 GPA: 4.0

- **Coursework:** MATH 531 Real Analysis; MATH 545 Stochastic Processes; MATH 501 Algebraic Structures I; MATH 403 Advanced Linear Algebra; MATH 221 Linear Algebra; MATH 281S Problem Solving Seminar; ECE 685D Intro to Deep Learning; ECE 689 Advanced Topics in Deep Learning; ECE 687D Theory & Algorithms of Machine Learning; COMPSCI 572 Natural Language Processing; Greek History; The Art of Improvising.

* Course numbers ≥ 500 (Duke) / ≥ 5000 (UMN) indicate graduate-level.

University of Minnesota, Twin Cities

Minneapolis, MN

Dual Enrollment (PSEO)

2022 – 2024 GPA: 4.0

- **Coursework:** CSCI 1133 Introduction to Programming Concepts; CSCI 2011 Discrete Structures; CSCI 1933 Introduction to Algorithms and Data Structures; CSCI 2041 Advanced Programming Principles (OCaml); CSCI 3003 Introduction to Computing in Biology; CSCI 4511W Introduction to Artificial Intelligence; CSCI 5521 Machine Learning Fundamentals; CSCI 5461 Functional Genomics, Systems Biology, and Bioinformatics; MATH 2373 Linear Algebra and Differential Equations; PHYS 1401V Honors Physics I (Top 10%); ME 3331 Thermodynamics; ME 3332 Fluid Mechanics.

University of Oxford, New College

Oxford, UK

Visiting Student: Ethics and Philosophy of Artificial Intelligence; *Portfolio Essay*

Summer 2025 Grade: A–

EXPERIENCE

Researcher at Yann LeCun's Lab

New York, NY

Computational Intelligence, Vision, and Robotics Lab at NYU

Feb 2025 – Present

- **LpWM:** a JEPA world model with non-negative sparse representations; showed sparsity lowers the predictor capacity needed for planning, beating dense world models by up to 57% on PushT.
- **RectifiedLpJEPA:** a controllable sparsity-based self-supervised learning method that trains models on maximum-entropy sparse representations, enabling highly sparse yet performant pretraining. See lpjepa.com for details.
- **Radial-VCReg:** a self-supervised learning method augmenting VICReg with a radial Gaussianization loss that transforms a broader class of feature distributions toward normality.
- Presented at the [ICML 2026 main conference](#), NeurIPS 2025 workshops ([Symmetry and Geometry in Neural Representations](#), [UniReps](#)), and [WM@Booth](#); [poster](#).

Member of Technical Staff Intern at Pantheon

San Francisco, CA

Robotics research & deployment lab solving general-purpose robotics; raised \$25M at a \$165M valuation

May 2026 – Aug 2026

- Researcher in residence working on JEPA-based world models for robot control; 5th hire at a company that has since grown to 30+ employees.
- Helped build out the research taste and direction of the founding team; worked on a first-author publication.

Research Engineer at Jiarong Hong's Lab

Minneapolis, MN

Flow Field Imaging Lab

2022 – 2024

- Led development of [SkyWindFarm](#), an airborne wind energy system; first author and inventor on a [utility patent](#).
- Applied Reynolds–Averaged Navier–Stokes with SST $k-\omega$ turbulence modeling to optimize vertical-axis wind turbines, improving the coefficient of performance (C_p) as a function of tip-speed ratio (λ).
- Conducted 6-DoF CFD simulations on the Agate supercomputer at UMN MSI, wind tunnel experiments at the Akerman Hall wind tunnel, and outdoor flying-prototype field tests to design, validate, and advance the system to TRL 4.
- Won \$30K+ in awards, including 3rd Place Grand Award at ISEF 2023 & 2024, \$10,000 Special Prize (ISEF 2024), and Regeneron Science Talent Search 2024 Finalist. Check out: [Project paper](#); [prototype flight video](#); [technical explanation](#).

PROJECTS

ConnectU.ai

2025

- Built [ConnectU.ai](#), a mentor–mentee pairing platform; formulated matching as a constrained optimization problem and implemented matching via the Hungarian algorithm over embedding similarity.

AI Systems for Distracted Driving Prevention

2022

- Designed a momentum-based safer-route planning algorithm over 3.1M+ accident records (22 years, Minnesota DOT); deployed live at [idontwannadie.lol](#) ([demo](#)); won Best Use of Statistics and Best Community Impact at PennApps XXV.
- Prototyped EyeDa, a real-time distracted-driving detection device, leading a 15-person engineering team; presented at the Minnesota Towards Zero Deaths Conference; covered by [KARE11](#), [CBS](#), and the [Star Tribune](#) ([device video](#)).

OTHER

Awards: [Regeneron STS Scholar](#); [YC AI SUS Hackathon — 1st Place](#); USNCO Qualifier (5th, MN); [National Merit Scholar](#)

Skills: Python, PyTorch, PyTorch Lightning, NumPy, Slurm, Stable Pre-training, Stable Datasets, Stable World Models, C/C++, JavaScript/TypeScript, React.js, Node.js, HTML/CSS, Java, SQL, MongoDB, Docker, Kubernetes, AWS, Julia, MATLAB, SolidWorks, Fusion 360, ANSYS Fluent.

Interests: Philosophy (Nietzsche, Girard), Mathematics, Reading, Art, Rowing, Running, Understanding and Building Intelligence.

Languages: English (Formally Proficient), Hindi (Formally Proficient), Love (Informally Proficient).